

Solid State Structural Aspects of Solid Self-microemulsifying Drug Delivery Systems (sSMEDDS) during Digestion

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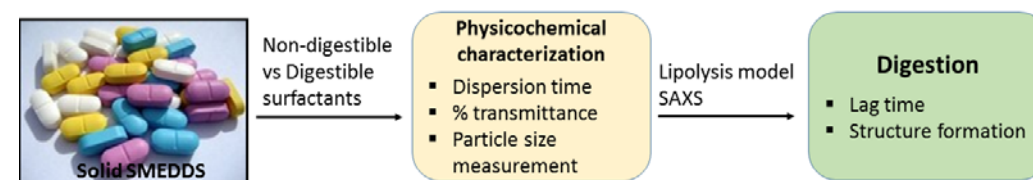
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PURPOSE

Solid self-microemulsifying drug delivery system (sSMEDDS) have received significant interest in recent times for delivering lipophilic drugs to overcome some drawbacks of liquid SMEDDS. The digestion and structure formation are essential aspects of such systems, knowledge of which enables understanding of drug solubilization and drug absorption. Therefore, the current study aimed to provide the first insight on the dynamic digestion process and structure formation phenomena in real-time of sSMEDDS containing of non-digestible (NDsur) and digestible surfactants (Dsur).



METHODS

- Pre-selected ratios (Table 1) of lipid and surfactants (w/w) were mixed and melted at 60°C on heating plate with magnetic stirrer. The melted liquid was solidified on a cylindrical mould at 4°C for 24 hours.

Table 1. Compositions of selected sSMEDDS formulations.

Formulations	Non-digestible (NDsur)			Digestible (Dsur)	
	Gelucire 44/14 (GEL)	Vitamin E TPGS (TPGS)	Lutrol® F 127 (F 127)	Sucrose esters S-1670 (S-1670)	Span® 60 (S-60)
GEL alone	100%	-	-	-	-
GEL40:NDsur ¹	40%	30%	30%	-	-
GEL50:NDsur	50%	25%	25%	-	-
GEL50:Dsur	50%	-	-	25%	25%

% ratios were in w/w

- Dispersion time, % transmittance and particle size were measured and thermal analysis was conducted using differential scanning calorimetry (DSC).
- The digestion process was performed using an *in vitro* lipolysis model at 37°C in a thermostatted glass reaction vessel which was connected to a pH stat apparatus. The formulations were pre-dispersed in 13.5 mL 'fasted' intestinal mixed micelles and digestion was initiated by adding 1.5 mL pancreatic lipase.

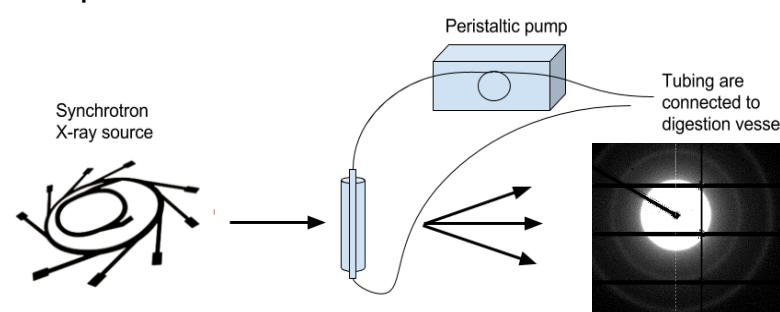


Figure 1. Small-angle X-ray scattering (SAXS) flow through set up at Australian synchrotron.

Table 2. Dispersion time and particle size distribution of sSMEDDS.

Formulations	Dispersion media	Dispersion time (min)	Small particles diameter (nm)	Small particles area (%)	Large particles diameter (nm)	Large particles area (%)
GEL alone	0.1 M HCl	9	21.1	15.8	330.6	78
	Digestion buffer	19	13.3	20.6	269.1	75
GEL40:NDsur	0.1 M HCl	7.69	18.9	89.6	366	9.8
	Digestion buffer	9.04	19.3	93.1	-	-
GEL50:NDsur	0.1 M HCl	13.33	30.4	60.5	277	39.5
	Digestion buffer	12.15	32.8	55.2	217	44.8
GEL50:Dsur	0.1 M HCl	20.16	24	8.7	222	89.4
	Digestion buffer	33.14	25.4	9.4	207	84

sSMEDDS containing of non-ionic hydrophilic surfactants, TPGS and F 127 displayed shorter dispersion times, favouring the formation of fine dispersed particles. The hydrophobic nature of S-60 enhanced the dispersion time with major particle size distribution from larger particles.

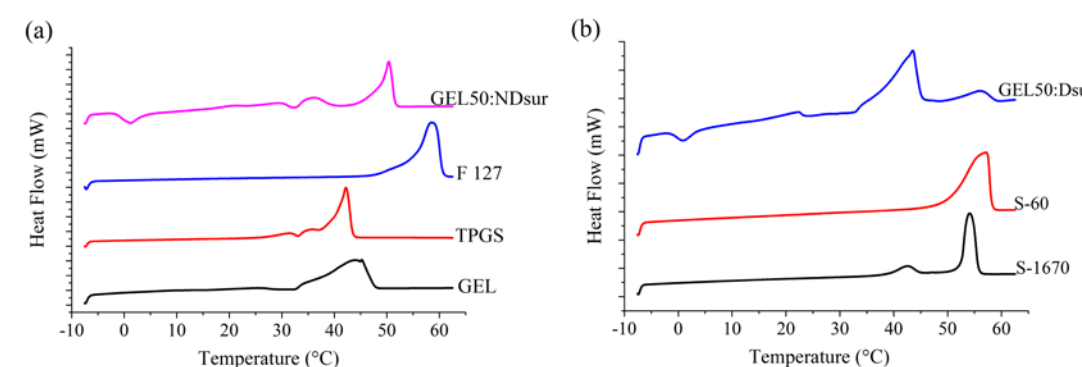


Figure 2. DSC thermograms of pure components and sSMEDDS.

Formulations indicated amorphous state similar to previous study¹. NDsur sSMEDDS indicated homogeneous mixing but S-1670 did not completely mixed in Dsur sSMEDDS.

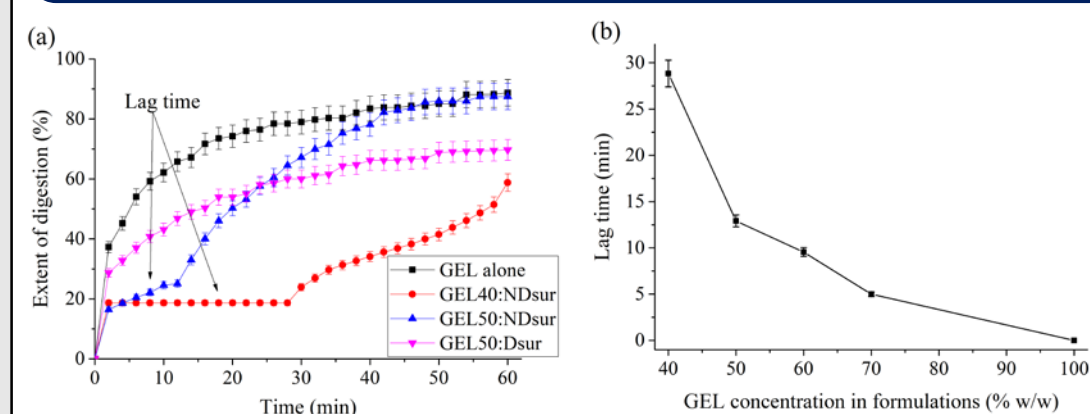


Figure 3. (a) Extent of digestion of GEL alone and sSMEDDS and (b) lag time behaviour of NDsur sSMEDDS, in fasted intestinal conditions over 60 min.

GEL alone digested 89% over 60 min. The inhibitory effect of TPGS² and non-digestible nature of surfactants exhibited GEL concentration dependent lag time. However, incorporation of Dsur instead of NDsur eliminated the lag time, indicating approximately 50% digestion in first 12 min after lipase addition.

RESULTS

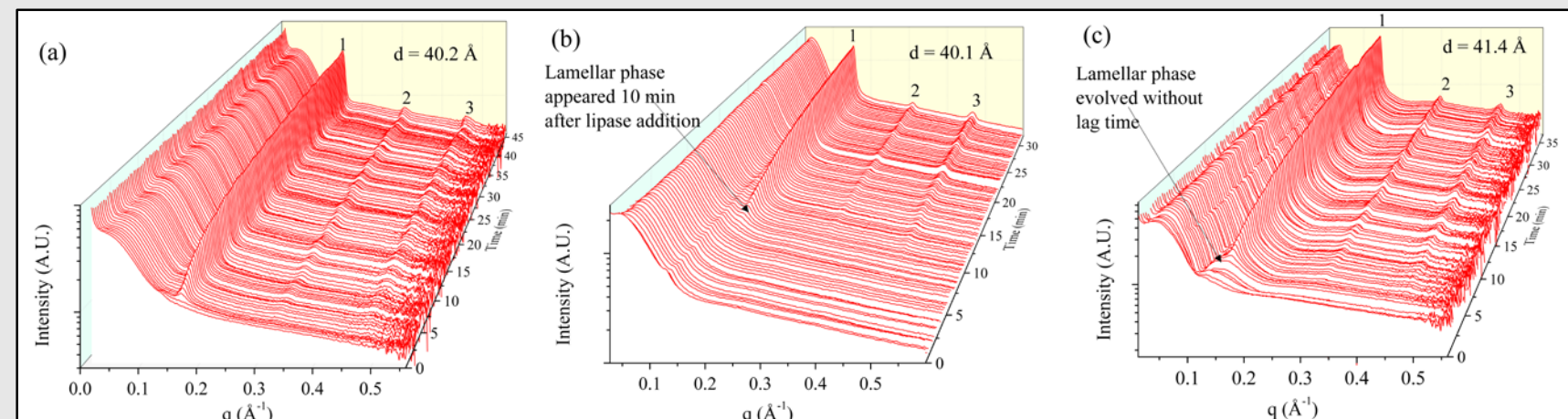


Figure 4. Time resolved structure evolution of (a) GEL alone, (b) GEL50:NDsur and (c) GEL50:Dsur during digestion in fasted intestinal micellar condition measured using synchrotron SAXS.

The lamellar phase was formed for GEL and sSMEDDS where the lag time was evident for NDsur.

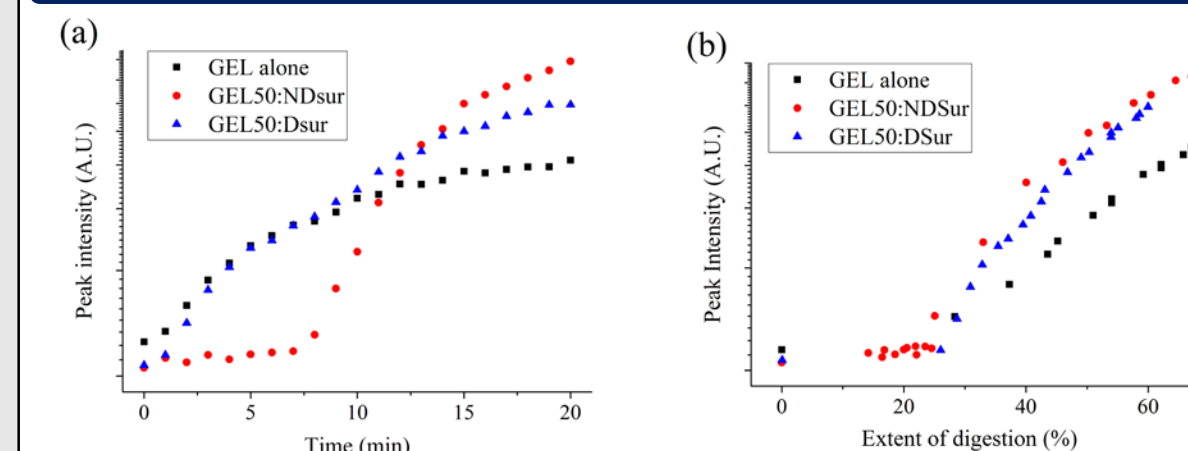


Figure 5. Correlation between the digestion kinetics from (a) digestion time and peak intensity and (b) extent of digestion and peak intensity of the main lamellar phase reflection at $q = 0.15 \text{ Å}^{-1}$

Peak intensity was in correlation with critical extent of digestion, not to time. A critical extent of digestion of approximately 25-30% was required before the formation of lamellar phase occurred.

CONCLUSIONS

- An optimum ratio of lipid to surfactants is prerequisite to develop rapidly digestible sSMEDDS.
- The kinetics of structure evolution is highly dependent on the main digestible component, regardless the types of surfactants.
- Future studies will address the impact of structure formation on fate of drug incorporated into sSMEDDS.

REFERENCES

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- A. Christiansen, T. Backensfeld, W. Weitschies. Effects of non-ionic surfactants on *in vitro* triglyceride digestion and their susceptibility to digestion by pancreatic enzymes. European Journal of Pharmaceutical Sciences, 41 (2), 376-382.

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